INDIAN ELECTRIC VEHICLES INDUSTRY

India's electric vehicle (EV) industry is entering a high-growth phase, fueled by rising fuel costs, environmental urgency, and aggressive policy support. With over 2 million EVs sold in 2024, the market is dominated by two-wheelers (E2Ws) and three-wheelers (E3Ws), which together account for 93% of total sales.

The government's vision for 30% EV penetration by 2030 is backed by flagship programs like FAME-II, PM E-DRIVE, and the PLI scheme, which collectively inject over ₹20,000 crore into infrastructure, manufacturing, and consumer incentives. States such as Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu, and Delhi lead adoption, while regions like Assam and Bihar present untapped potential.

⚙️ Policy & Infrastructure Momentum

- FAME-II Phase II: ₹10,000 crore allocated to subsidize EV purchases and expand charging infrastructure
- PM E-DRIVE Scheme: ₹10,900 crore to accelerate EV ecosystem development
- Customs Duty Exemptions: On 25 critical minerals including lithium and cobalt for battery production
- Charging Infrastructure Goal: 1.32 million public charging stations by 2030

📈 Market Growth & Investment Outlook

- EV market projected to grow from US$ 3.2 billion in 2022 to US$ 113.9 billion by 2029, at a 66.5% CAGR
- Battery demand reached ~13 GWh in 2024, with 3Ws and 4Ws leading consumption
- Tamil Nadu alone expects ₹50,000 crore in EV investments by 2025, creating ~150,000 jobs
- Indian automakers plan to launch over a dozen new EV models in 2025, focusing on premium and fleet segments

🌍 Sustainability & Global Positioning

India is positioning itself as a global hub for EV innovation and auto component sourcing, with over 25% of production exported annually. The Bharat Mobility Global Expo 2025 signals India's ambition to lead in clean mobility, targeting ₹8.65 lakh crore in exports through cost-competitive manufacturing and tech leadership.

🚀 Innovation & Technology Landscape

India's EV ecosystem is rapidly evolving beyond just vehicle sales. The rise of battery-as-a-service, swappable battery models, and connected EV platforms is reshaping how consumers interact with mobility. Startups and OEMs are investing in AI-driven fleet management, IoT-enabled diagnostics, and smart charging networks, making EVs not just sustainable but intelligent.

Key players like Tata Motors, Ola Electric, Ather Energy, and Mahindra are integrating advanced telematics, over-the-air updates, and predictive maintenance into their EV offerings. Meanwhile, battery tech is shifting toward solid-state chemistry, LFP cells, and localized recycling, with India's battery demand projected to hit 27.7 GWh by 2028.

🗺️ Regional Dynamics & Adoption Patterns

- Uttar Pradesh leads in volume, driven by e-rickshaws and last-mile mobility
- Maharashtra & Karnataka show strong urban adoption of premium 2Ws
- Tamil Nadu is emerging as a manufacturing and export hub, with ₹50,000 crore in EV investments expected by 2025
- Assam & Bihar, despite low sales, represent high-potential markets for affordable EVs and rural electrification

State-level EV policies vary widely, with some offering road tax waivers, registration subsidies, and charging infra mandates, while others lag in execution. This uneven policy landscape creates both challenges and opportunities for OEMs and fleet operators.

🕰️ History of the Indian Electric Vehicle Industry

🔹 1990s: Early Experiments
- Small electric scooters and three-wheelers emerge to combat urban pollution
- Infrastructure and battery tech were underdeveloped; adoption remained niche

🔹 1996: Reva Electric Car Launched
- Developed by Maini Group in Bengaluru
- India's first electric car, designed for short city commutes
- Gained global attention but limited domestic traction

🔹 2001–2010: Dormant Phase
- Reva acquired by Mahindra in 2010
- EVs remained limited to prototypes and pilot programs
- Lack of incentives and charging infra stalled growth

🔹 2010: First Government Push
- Ministry of New and Renewable Energy introduces EV subsidy scheme
- Focused on boosting domestic production and consumer adoption

🔹 2013: National Electric Mobility Mission Plan (NEMMP)
- Strategic roadmap to reduce oil dependency and improve air quality
- Target: 6–7 million EVs by 2020 (partially achieved)

🔹 2015: FAME India Scheme (Phase I)
- Financial incentives for EV buyers and infrastructure development
- Focused on two-wheelers, three-wheelers, and public transport

🔹 2019: FAME II Launch
- ₹10,000 crore allocated
- Expanded support for electric buses, charging stations, and fleet electrification
- Catalyzed OEM investment and startup activity

🔹 2021–2023: Market Acceleration
- EV sales cross 1 million units annually
- Rise of Ola Electric, Ather, Tata Nexon EV, and BYD
- State-level policies and subsidies gain traction

🔹 2024: EV Sales Cross 2 Million Units
- Dominated by E2Ws (81%) and E3Ws (16%)
- Charging infra expands to 10,000+ public stations
- Over ₹3 billion raised in EV investments

🔹 2025: Bharat Mobility Global Expo & PM E-Drive Scheme
- India positions itself as a global EV hub
- New policy replaces FAME with broader ecosystem support
- Focus on exports, battery recycling, and carbon-neutral manufacturing

EV MAKERS GEOGRAPHICAL DETAILS

CLEAR SLICERS

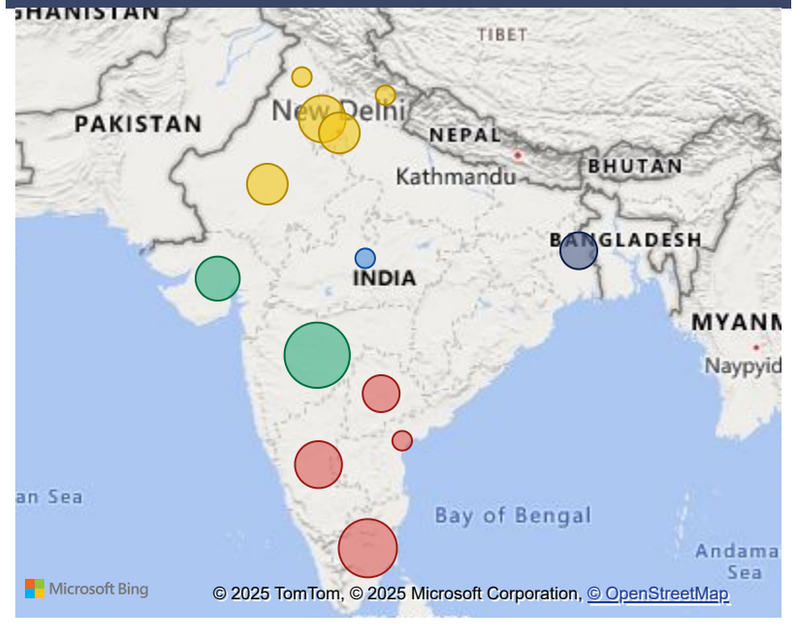
Cluster: All

State: All

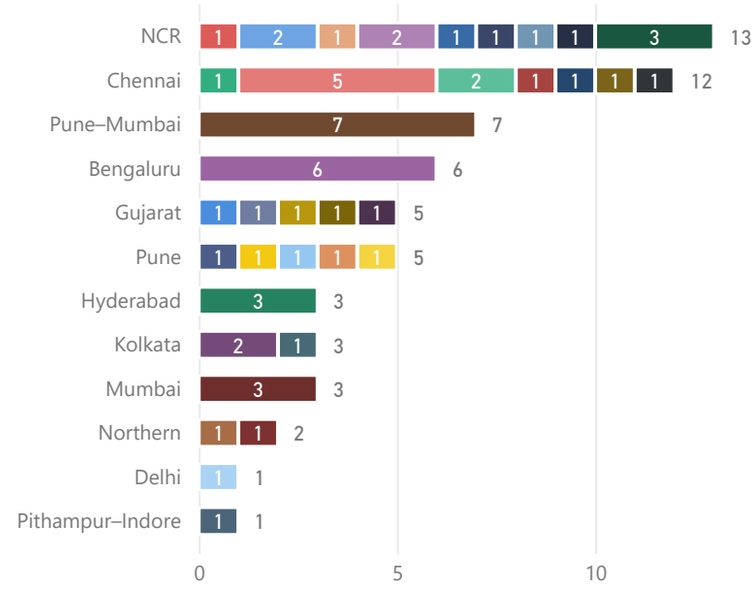
City: All

EV Maker: All

EV Makers by State and Region



GEO_State V OperationalPCS



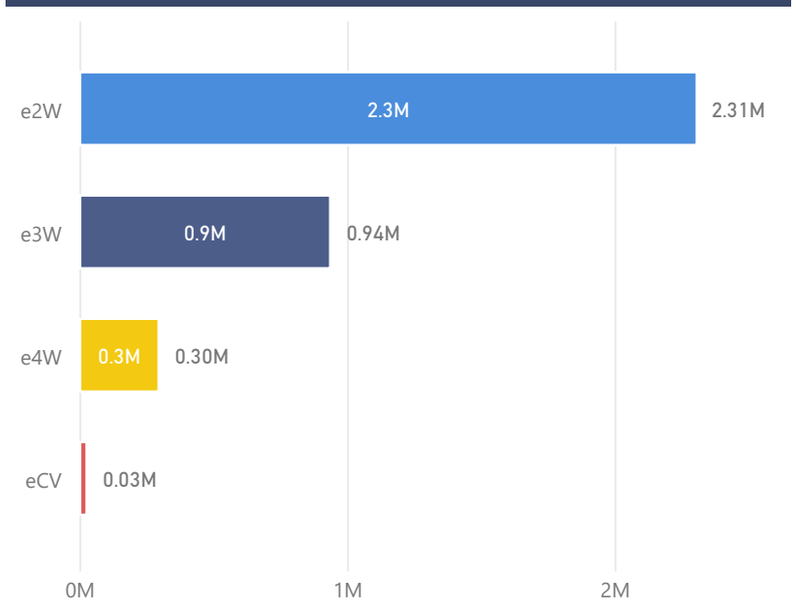
EV Makers Locations

State	City	EV Maker
Gujarat	Ahmedabad	Electrotherm India
Maharashtra	Ahmednagar	Kinetic Green (3-Wheeler)
Maharashtra	Akurdi	Bajaj Auto (EV)
Rajasthan	Alwar	Ashok Leyland (EV Bus)
Andhra Pradesh	Anantapur	Kia Motors India
Maharashtra	Aurangabad	Audi India
Maharashtra	Baramati	Piaggio Vehicles
Karnataka	Bengaluru	Altigreen Propulsion Labs
Karnataka	Bengaluru	Ather Energy
Karnataka	Bengaluru	Bosch India
Karnataka	Bengaluru	Mahindra Electric
Karnataka	Bengaluru	Mahindra Reva
Karnataka	Bengaluru	Simple Energy
Rajasthan	Bhiwadi	Okinawa Autotech
Rajasthan	Bhiwadi	Saera Electric Auto
Gujarat	Bhuj	Triton Electric

Segment: All

Manufacturer: All

GEO_State V OperationalPCS



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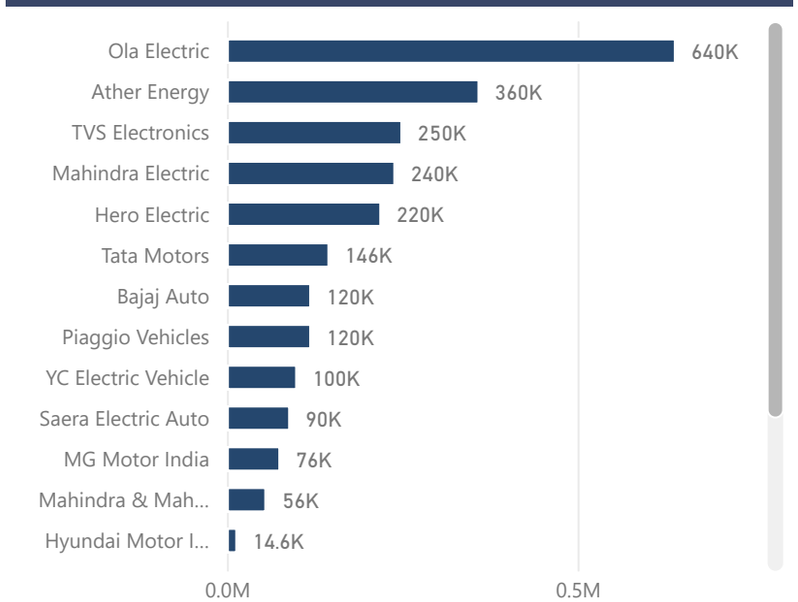

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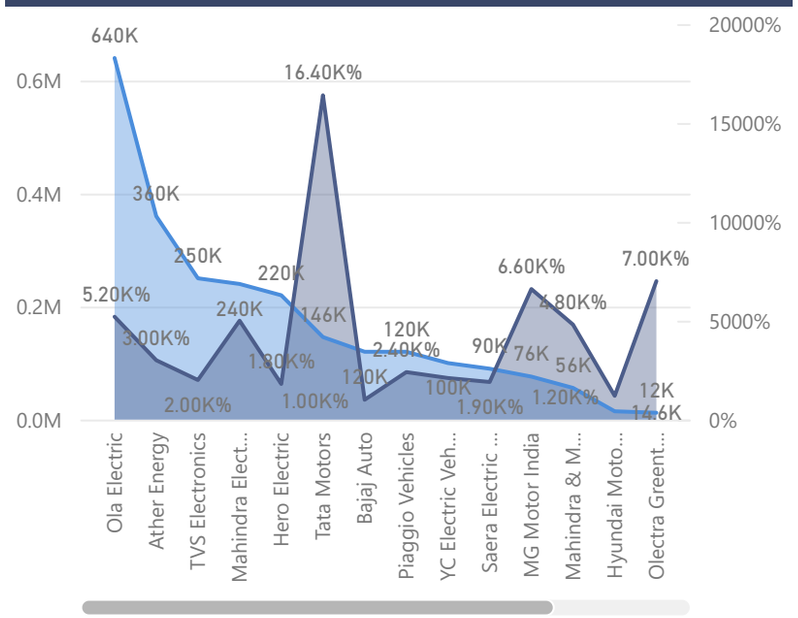


SALES ANALYSIS BY SEGMENT

CLEAR SLICERS

INDIAN ELECTRIC VEHICLES INDUSTRY

Sales vs Margin Sales	YoY Rate	Prime_SalesValue
435.70✓	15.23% !	2K
Goal: 378.87 (+15%)	Goal: 16.00 (-99.05%)	
2025	2025	

GEO ANALYSIS

PRODUCTION

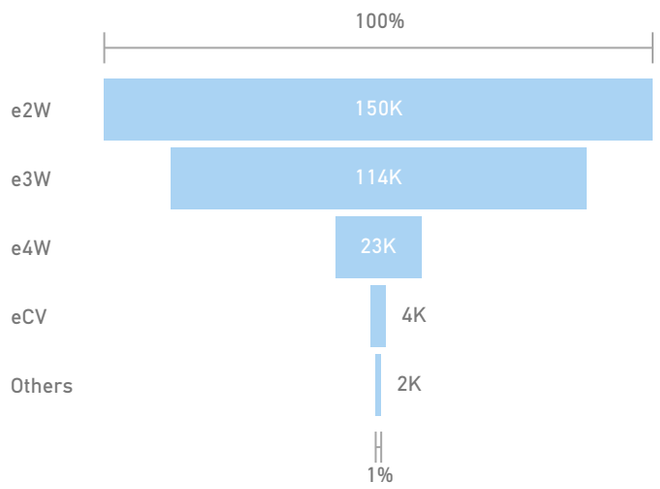
PARETO ANALYSIS

- INTRODUCTION
- HISTORY & GEOGRAPHY
- SALES
- INDUSTRY STATUS
- CHALLENGES & GOVT POLOCIES
- FUTURE SCOPE

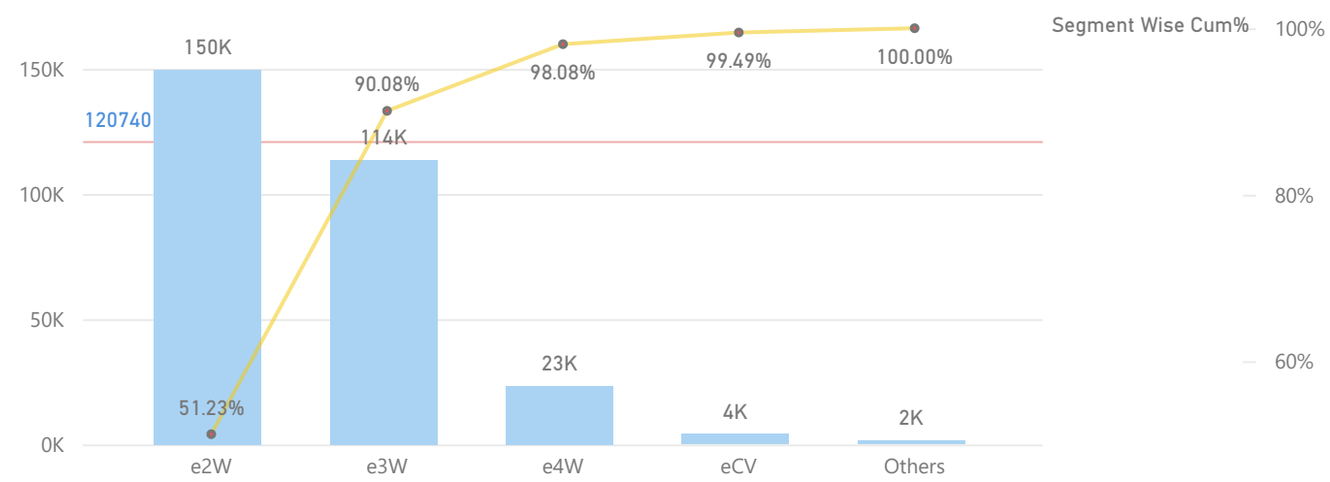
Year

2023

SEGMENT WISE SALES



TOP 80% SALES VALUE PROVIDING SEGMENTS

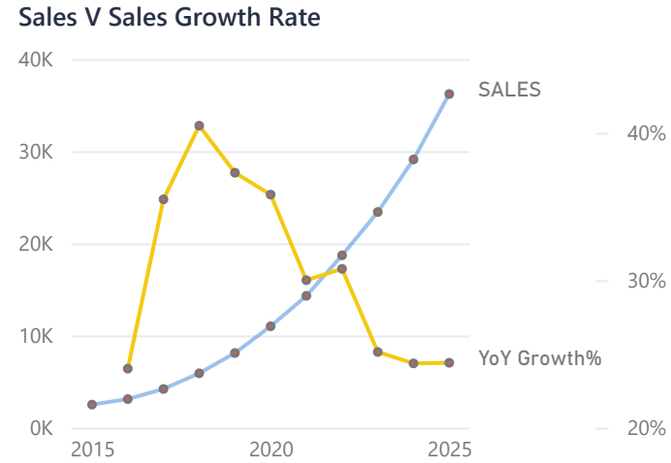


TIME ANALYSIS

Segment

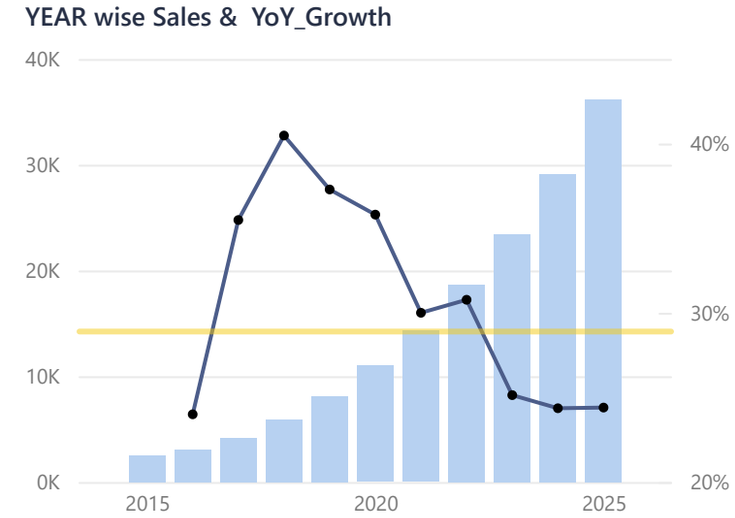
e4W

Sales V Sales Growth Rate



Fiscal Year	Sales	Cum%
2015	2500	2.27%
2016	3100	5.12%
2017	4200	8.74%
2018	5900	13.38%
2019	8100	19.36%
2020	11000	27.03%
2021	14300	36.67%
2022	18700	48.81%
2023	23400	63.41%
2024	29100	80.46%
2025	36200	100.00%
Total	**156500**	

YEAR wise Sales & YoY_Growth



INDIAN ELECTRIC VEHICLES INDUSTRY

SALES ANALYSIS BY SALES ANALYSIS BY GEOGRAPHY

CLEAR SLICERS

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2025	2025	

GEO ANALYSIS

PRODUCTION

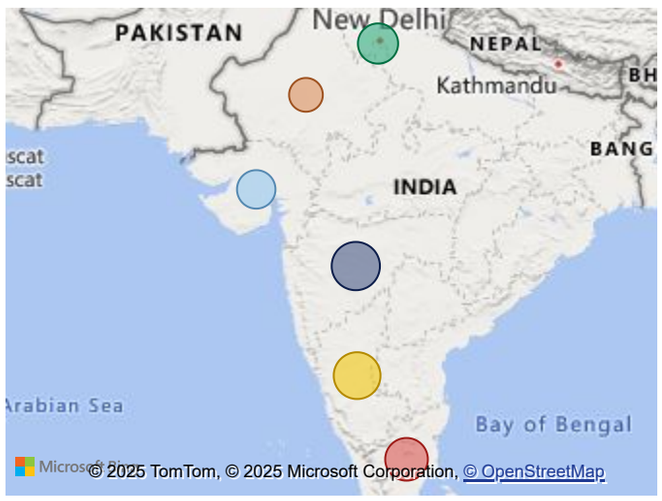
PARETO ANALYSIS

Cluster: All

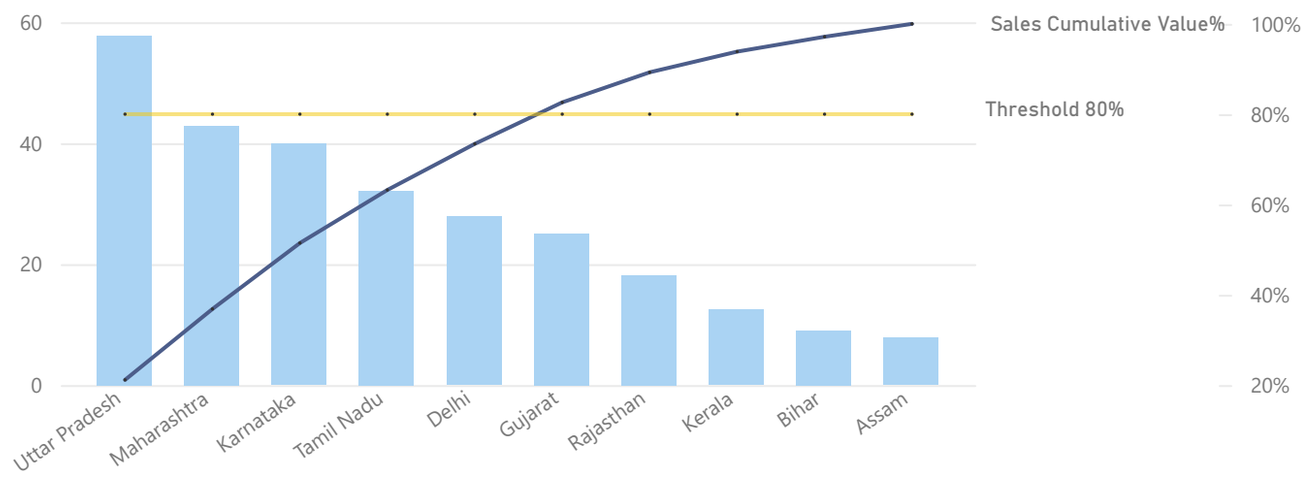
State: All

Year: 2022

STATE WISE SALES



TOP 80% SALES VALUE PROVIDING SALES



PESTLE ANALYSIS

STATES: All

Concern Type: All

States +/- Comparision (Values)

State	Charging networks	Commercial EV licensing	cultura
Maharashtra	1	0	
Karnataka	0	0	
Tamil Nadu	0	0	
Delhi	0	0	
Gujarat	1	1	
Rajasthan	0	0	

States +/- Analysis (Concerns)

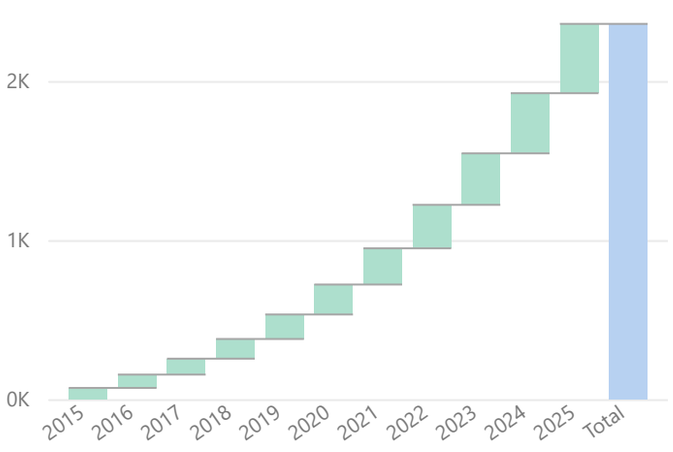
Value	0		1	
State	Concern 1	Concern 2	Concern 1	Concern 2
Maharashtra	Commercial EV licensing	sustainability goals	Charging networks	subsidies
Karnataka	Charging networks	sustainability goals	industrial EV push	stable governance
Tamil Nadu	Charging networks	shared mobility	EV-friendly registration	sustainability goals
Delhi	Charging networks	sustainability goals	EV-friendly registration	subsidies
Gujarat	cultural inertia	sustainability goals	Charging networks	stable governance

** here +'1' represent Higher Capability , '0' for Medium Capability and -'1' for Poor Capability

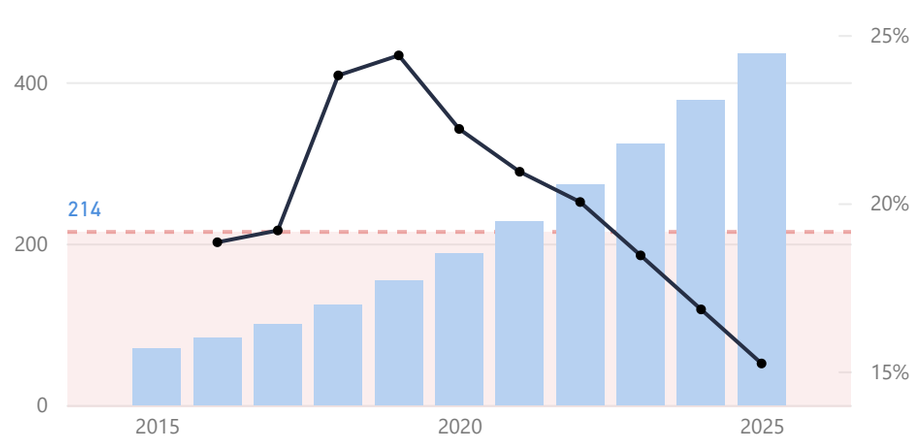
TREND ANALYSIS by YEAR

STATE: All

SALES TREND by YEARS



STATE WISE SALES

State	Assam	
Year	UnitSold	YoY_Grow
2015	2	
2016	3	20.
2017	4	20.
2018	4	20.
2019	5	21.
2020	6	17.
2021	7	15.
2022	8	13.
2023	9	11.
2024	10	10.
2025	11	9.

SALES TREND & YoY GROWTH